

INTRODUCTION

During times of crisis, news can share some triggering and foresightful information. Because in a country approaching a crisis, crises don't happen suddenly; they develop gradually, based on certain movements. In this project, I wondered if we could actually predict crises by taking insights from news. To investigate this, I chose two countries: Turkey and Argentina, both of which experienced crises in 2018. I collected news headlines and then, using an AI model, labeled these headlines as positive, negative and neutral. Positive and neutral news doesn't predict crises but negative news points to a financial problem beforehand. My main focus in this project is these negative marks. In my research, the two countries showed different results. In Turkey, negativity was very high before the crisis but in Argentina it increased more after the crisis. Although Turkey's situation was as I predicted, Argentina's situation completely overturned my predictions.

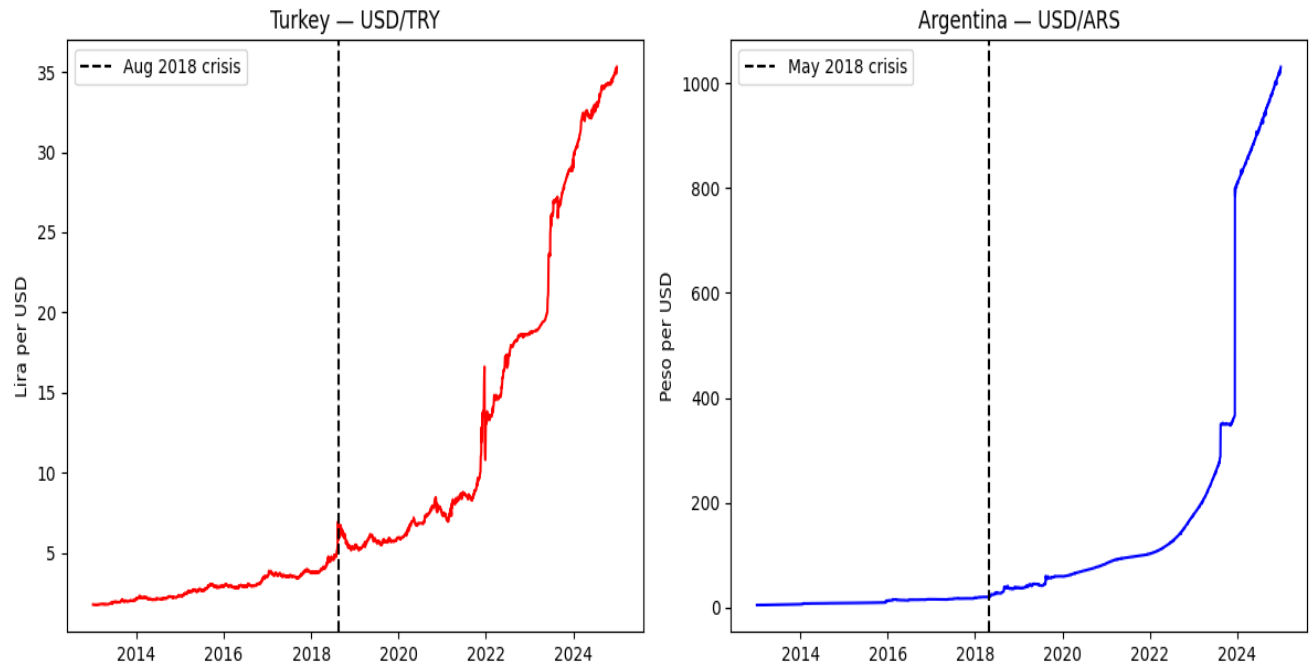
LITERATURE

Tetlock (2007) measured the daily tone of a newspaper. He concluded that the more pessimistic the news in the newspaper, the more the stock market fell but then prices returned to their previous levels. From this, it was concluded that people's emotions change the direction of the stock market. I apply the same logic but not to the stock market but to the currency crises of developing countries. When negative reports in global news increase, capital tends to leave the country. But the difference in my project is that, with the advancement of artificial intelligence today, I do this word by word with the help of an artificial intelligence model instead of my own analysis.

Data

I collected the news data via the gdelt API. The gdelt API has data covering a very long period up to the present day. I used this API to collect this data for Turkey and Argentina. Of course, I collected all the news only the period I needed. Specifically, I applied this to both countries, 3 months before the crisis and 3 months after the crisis and as a result: I collected 1500 news items for Turkey and 1456 for Argentina. In addition, I needed exchange rate data. Because I was working on the currency crisis, I needed to see currency changes both the actual date the crisis erupted and how much impact these changes had by correlating them with the dates of the news reports. I pulled the currency data from Yahoo Finance; this time I used a data site instead of an API.

Note: When I started the assignment, I initially thought of three countries: Russia, Turkey, and Argentina. However, the gdelt API constantly gave me rate limits. I used other methods for a few days, and I was able to retrieve the data for Turkey and Argentina from the API over several days by using a different IP address each day. Therefore, if I had applied the same logic to Russia, I wouldn't have been able to complete the assignment, so I removed Russia.



This graph shows the exchange rates of various countries against the dollar. The black lines indicate crisis dates. I determined these dates by looking at the largest daily exchange rate jumps.

Methodology

I divided the news into three categories: negative, neutral, and positive. However, I didn't do this using readily available data or on my own; I did it with the help of an LLM. Specifically, thanks to the prompt I gave the AI, it can now understand which category a text belongs to and group them accordingly. I had over 3000 news articles. Gemini's limit was 500. So I selected 480 articles and performed an analysis on them. In my first selection, due to inexperience, the vast majority of the articles were related to Turkey. Python, instead of randomly mixing them almost exclusively selected articles from Turkey. I then edited it and ended up with 480 completely random articles with an equal number of articles for each country.

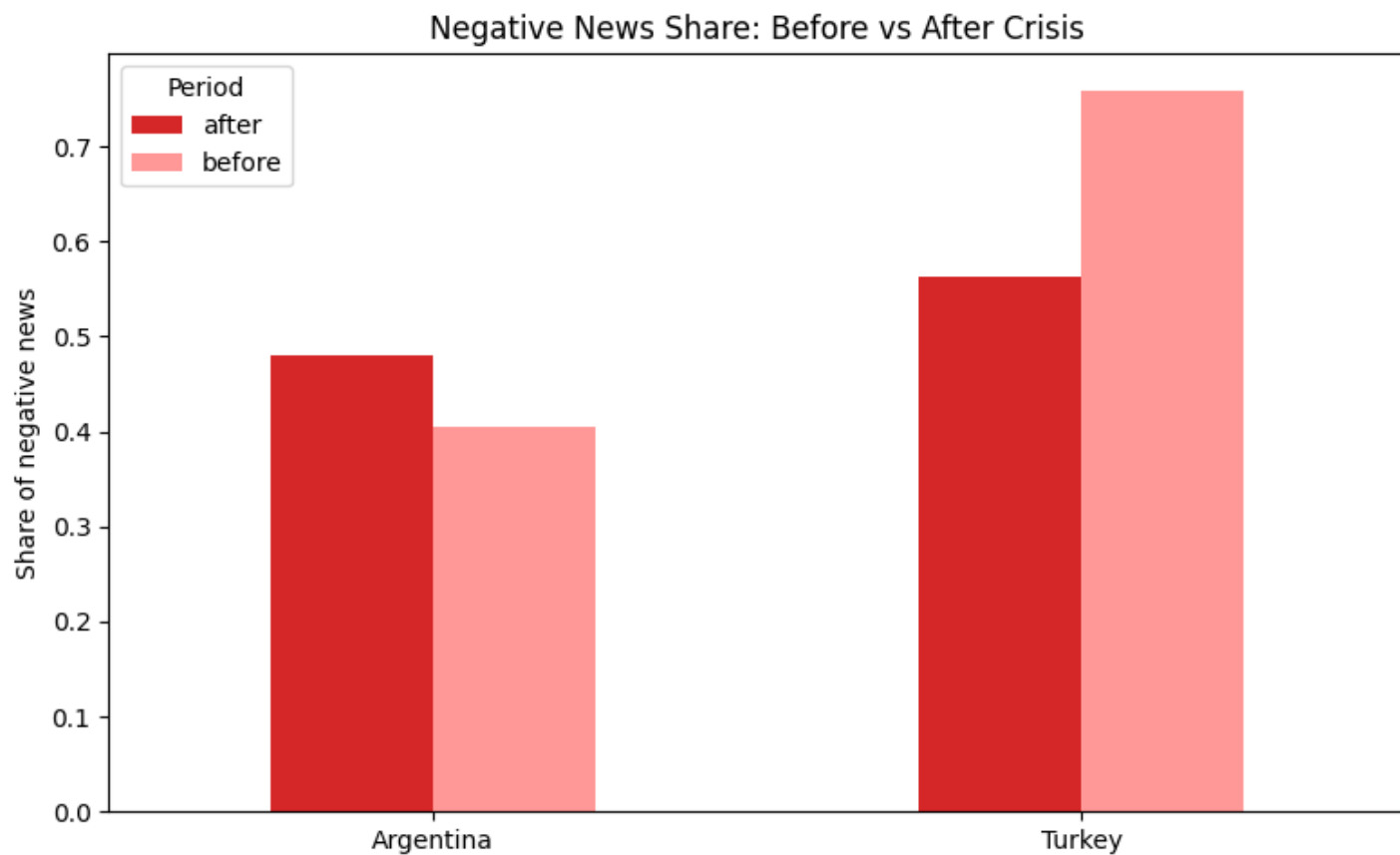
Next, I extracted the selected news items for each country from the list along with their

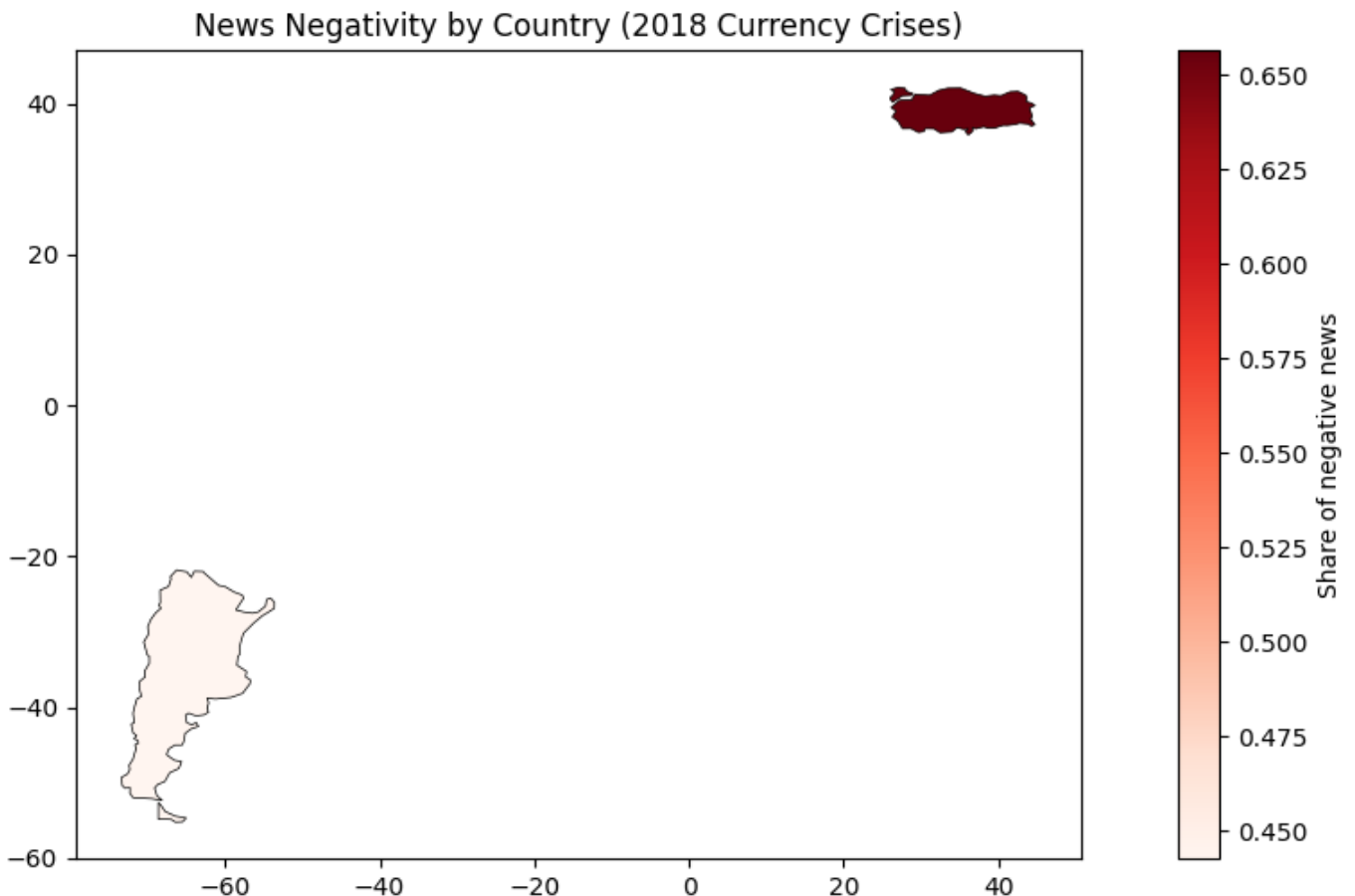
dates. Then, I divided these news items into pre crisis and post crisis categories allowing me to perform the analysis. For the analysis, I created a graph using both a bar chart and the spatial data of the countries.

Results

Looking at the data, I encountered a problem: Turkey's news coverage (65.6%) is more negative than Argentina's (44.3%). I concluded that this is actually logical because Turkey deals more with sudden crises. I've observed that crises in Turkey come suddenly and last for a period. Argentina's crisis history is different in here. There are continuous, long term crises.

Now, my main analysis is can news reports confirm a crisis before it happens? Here, I encountered a different picture than I expected. In Turkey, it turned out as I thought: the negativity rate in news reports before the crisis was 75.9% while after the crisis it was 56.3%.





So, as I expected, news reports before the crisis were more negative and it was actually possible to glean information about Turkey's crisis period from the news. But in Argentina it was the opposite. When we look at Argentina, we see that the negativity rate, which was 40.5% before the crisis, rose to 48% after the crisis. Although, I don't know the exact reason why this is so unusual for Argentina from what I can see, there are so many crises that none of them have any particular significance and crisis news isn't really in the spotlight anymore.

However, my current project cannot definitively prove that there were signals in the news before the crisis. Firstly, the two countries are very few, and the analysis I'm using is limited to only 480 news items. Therefore, I need more news and data to make such a claim.

Conclusion

In this project, I investigated whether we can predict currency crises in advance using news reports. My findings indicate that it's indeed possible to confirm crises in the case of Turkey but not so much in Argentina due to the constant negative news. This project could be expanded in the future to include more countries such as developing countries and research could be conducted using thousands of news data points.

Usage of ai

Throughout the entire project, I only used Claude in one chat, and from the beginning of the chat, I asked him what he was telling me, and I copied his answer directly here.

Here's where I came in during the project:

1. When IMF and FRED didn't work, I suggested Yahoo Finance and GDELT as alternative data sources.
2. I explained GDELT's query syntax — specifically the `sourcelang:english` filter for restricting results to English-language articles.
3. I helped design the sentiment classification prompt — the instruction telling the model to label each article as positive, negative, or neutral.
4. I wrote the matplotlib code for the exchange rate chart.
5. I suggested shuffling the sample to correct the country imbalance.
6. I helped with the choropleth and bar chart styling — the `column=`, `cmap=`, and `legend` parameters.
7. I explained how to merge `seendate` back in by matching on the article titles.
8. I gave you the GDELT datetime format string (`%Y%m%dT%H%M%S`).

Reference:

Tetlock, P. C. (2007). Giving Content to Investor Sentiment: The Role of Media in the Stock Market. The Journal of Finance